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| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO. | CONFIRMATION NO. |
|---|---|---|---|---|
| 18/416,753 | 01/18/2024 | Yu Fang | SONA-0015-U01-C01 | 9685 |

87084 7590 08/27/2024

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

| EXAMINER | |
|---|---|
| ALGIBHAH, HAMZA N | |

| ART UNIT | PAPER NUMBER |
|---|---|
| 2441 | |

| NOTIFICATION DATE | DELIVERY MODE |
|---|---|
| 08/27/2024 | ELECTRONIC |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

| Office Action Summary | Application No. 18/416,753 | Applicant(s) Fang et al. | |
|---|---|---|---|
| | Examiner HAMZA N ALGIBHAH | Art Unit 2441 | AIA (FITF) Status Yes |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☑ Responsive to communication(s) filed on 07/25/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims***

5) ☑ Claim(s) 1-8,10-16 and 18-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-8,10-16 and 18-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

**Application Papers**

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 01/18/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

**Detailed Action**

1. ***Claims 1-8, 10-16 and 18-19*** are pending.

2. ***Claims 1-8, 10-16 and 18-19 are rejected.***

***Claim Rejections - 35 USC § 101***

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

3. Claims ***1-8, 10-16 and 18-19*** are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more. Claim 1 recites interpreting a vehicle network performance description; determining a zonal architecture value in response to the vehicle network performance description and a vehicle performance network impact description; and installing a multi-zone network on a vehicle in response to the zonal architecture value .

The above limitations correspond to a process, that under its broadest reasonable interpretation, covers "certain methods of organizing human activity" and/or performance of the limitations in mind. That is, nothing in the claim elements precludes the step from practically being "certain methods of organizing human activity" and/or being performed in mind. For example, "interpreting" in the context of this claim encompasses the user manually taking/understanding a vehicle network performance description. Similarly, the limitation of "determining a zonal architecture value" in the context of this claim encompasses the user manually calculating a zonal architecture value. Similarly, the limitation of "installing a multi-zone network" in the context of this claim encompasses the user manually installing physical network components onto a vehicle. In addition, the above limitations can be performed in human mind using a pen and a paper to perform interpreting, determining and installing. If a claim limitation, under its broadest reasonable interpretation, covers performance of the

limitation in the mind and/or covers "certain methods of organizing human activity" but for the recitation of generic computer components, then it falls within the "Mental Processes" and/or covers "certain methods of organizing human activity" grouping of abstract ideas. Accordingly, the claim recites an abstract idea.

This judicial exception is not integrated into a practical application because it does not impose any meaningful limits on practicing the abstract idea. The claim is directed to an abstract idea.

The claim(s) include additional element "at least one processor". However this element is not sufficient to amount to significantly more than the judicial exception. The courts have found that some computerized systems implementing abstract ideas do not add significantly more to the abstract idea. Thus, providing at least one processor does not automatically satisfy a 'particular machine' for this analysis. The recitation of the processor limitations amounts to mere instructions to implement the abstract idea on a processor, such as using at least one processor to interpreting a vehicle network performance description and determining a zonal architecture value. Taking the elements both individually and as a combination, the processor component at each step of the process perform purely generic computer functions. The claim as a whole does not amount to significantly more than the abstract idea itself.

Claims 2-8 and 10-11 depend on claim 1 and do not include additional elements that are sufficient to amount to significantly more than the judicial exception because all of the claimed limitations are part of the abstract idea.

Claim 12 recites interpreting a plurality of vehicle network performance descriptions; determining a zonal architecture value in response to the plurality of vehicle network performance

descriptions and a vehicle performance network impact description; and installing a multi-zone network on a each of a first vehicle and a second vehicle in response to the zonal architecture value.

The above limitations correspond to a process, that under its broadest reasonable interpretation, covers "certain methods of organizing human activity" and/or performance of the limitations in mind. That is, nothing in the claim elements precludes the step from practically being performed in the mind. For example, "interpreting" in the context of this claim encompasses the user manually taking/understanding a plurality of vehicle network performance descriptions. Similarly, the limitation of "determining a zonal architecture value" in the context of this claim encompasses the user manually calculating a zonal architecture value. Similarly, the limitation of "installing a multi-zone network" in the context of this claim encompasses the user manually grouping/combining a multi-zone network. If a claim limitation, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic computer components, then it falls within the "Mental Processes" grouping of abstract ideas. Accordingly, the claim recites an abstract idea.
This judicial exception is not integrated into a practical application because it does not impose any meaningful limits on practicing the abstract idea. The claim is directed to an abstract idea. The claim(s) does/do not include additional elements that are sufficient to amount to significantly more than the judicial exception because all of the claimed limitations are part of the abstract idea. The claim is not patent eligible.

Claims 13-16 and 18-19 depend on claim 1 and do not include additional elements that are sufficient to amount to significantly more than the judicial exception because all of the claimed limitations are part of the abstract idea.

***Claim Rejections - 35 USC § 102***

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

4. ***Claims 1-8 and 10-11 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by CHEVALIER et al (Pub. No.: US 2024/0031439 A1).

***As per claim 1, CHEVALIER discloses a method, comprising:***

- ***interpreting, via at least one processor, a vehicle network performance description that comprises a hardware redundancy description*** (CHEVALIER, paragraph 0052, 0132 wherein the points or locations at which a redundancy arrangement (e.g. triple modular redundancy) is provided can be determined by means of a fault mode analysis (FMECA). This fault/error mode analysis may allow the identification of critical components or paths within the network based on the selected allowed fault tolerance (e.g. single point failures, double point failures, triple point failures, etc.). Based on the result of the fault mode analysis, some selected vertices in the network are arranged in a redundancy arrangement (e.g. triple modular redundancy)) ***for a multi-zone network*** (CHEVALIER, Fig 7, paragraph 0064, 0183, wherein he network has a primary wheel topology arrangement and a secondary wheel topology arrangement, wherein the redundant subsets are connected in the primary wheel topology arrangement, and wherein the non-redundant subsets are connected in the secondary wheel topology arrangement)***;***
- ***determining, via the at least one processor, a zonal architecture value in response to***

***the vehicle network performance description and a vehicle performance network impact description that comprises a reliability description for the multi-zone network*** (CHEVALIER, Fig 7, paragraph 0053, 0097, wherein some selected vertices in the network are arranged in a redundancy arrangement (e.g. triple modular redundancy). The reliability of each of the components can be analyzed to determine a failure rate (e.g. mean time between failure or the like). From such results it can be monitored which components are sensitive in the moving machine and which are to be protected by applying a redundancy arrangement in order to reduce the failure rate of the moving machine)***; and***

- ***installing a multi-zone network on a vehicle in response to and configured in accordance with the zonal architecture value*** (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology).

As pre claim 2, claim 1 is incorporated and CHEVALIER discloses wherein the vehicle network performance description comprises at least one of the following: a latency value between at least two end points for the multi-zone network; a bandwidth value between at least two end points for the multi-zone network; or a control redundancy description for the multi-zone network (CHEVALIER, paragraph 0132, wherein the combination of such software

techniques with the implemented hardware redundancy arrangements can further increase the reliability of the cyber-physical system of the dump truck and improve the overall availability of the dump truck to the mining haulage process).

As pre claim 3, claim 1 is incorporated and CHEVALIER discloses wherein the vehicle performance network impact description comprises at least one of the following: an installed hardware cost value for the multi-zone network; a mission capability description for the multi-zone network; or a redundancy capability description for the multi-zone network; (CHEVALIER, Fig 7, paragraph 0053, 0097, wherein some selected vertices in the network are arranged in a redundancy arrangement (e.g. triple modular redundancy). The reliability of each of the components can be analyzed to determine a failure rate (e.g. mean time between failure or the like). From such results it can be monitored which components are sensitive in the moving machine and which are to be protected by applying a redundancy arrangement in order to reduce the failure rate of the moving machine).

As pre claim 4, claim 1 is incorporated and CHEVALIER discloses wherein the zonal architecture value comprises a network topology description (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability

items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology).

As pre claim 5, claim 4 is incorporated and CHEVALIER discloses wherein the zonal architecture value further comprises a zone manager distribution description (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology)..

As pre claim 6, claim 5 is incorporated and CHEVALIER discloses wherein the zonal architecture value further comprises a network type value for each zone of the multi-zone network (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology).

As pre claim 7, claim 6 is incorporated and CHEVALIER discloses wherein the zonal architecture value further comprises a network protocol value for each zone of the multi-zone network (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology).

As pre claim 8, claim 1 is incorporated and CHEVALIER discloses wherein the zonal architecture value further comprises a specification value for at least one hardware component of the multi-zone network (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology).

As pre claim 10, claim 1 is incorporated and CHEVALIER discloses wherein the multi-zone network is configured in accordance with a network topology description of the zonal architecture value (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology).

As pre claim 11, claim 1 is incorporated and CHEVALIER discloses wherein the multi-zone network is configured in accordance with a zone manager distribution description of the zonal architecture value (CHEVALIER, Fig 7, paragraph 0097, wherein the invention provides for a method of arranging a network of a cyber-physical system for a vehicle capable of autonomous or semi-autonomous moving, the method comprising the steps of receiving an initial network design with a plurality of interconnected distributed units, wherein the plurality of units includes sensors, actuators, and vertices (e.g. embedded systems); performing a fault analysis to identify lower reliability items in the initial network design with a reliability lower than a threshold value; arranging the lower reliability items in redundancy arrangements; interconnecting the redundancy arrangements in a fault tolerant network topology).

***Conclusion***

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441